BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . MEDMASTERS COLLABORATIVE

Module 6: Skeletal System

Week 6 teaching guide . 4 topics

This is the instructor-facing companion to the student pre-work hub. For every topic, you have the science to teach, the recommended approach to learning that material, and why it matters to your students' future patients. Lecture order, common misconceptions, and pre-video drawing prompts are baked in so each video lesson can hit the same pedagogical beats.

Topics in this module

Module 6 of 17 . Dr. Sharilyn Rennie

TOPIC 1 OF 4 . WEEK 6 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Bone Tissue and Bone Growth

Cells of bone, compact vs spongy, intramembranous vs endochondral ossification, and lifelong remodeling.

The Science

Bone is connective tissue with a mineralized matrix. It is alive and constantly being remodeled. The cells are osteoblasts (build matrix), osteocytes (mature, trapped in lacunae, sense load), and osteoclasts (resorb matrix; derived from monocyte lineage, not osteoblasts). Osteoblast versus osteoclast balance determines whether bone is being added or removed.

Compact bone is dense, organized into osteons (Haversian systems). Each osteon has concentric lamellae of matrix around a central canal carrying vessels and nerves. Spongy bone is the trabecular meshwork in the interior of bones, often containing red marrow. Long bones have a diaphysis (shaft), epiphyses (ends), metaphysis (growth zone), periosteum (outer covering), and endosteum (lining the marrow cavity).

Two kinds of ossification: intramembranous (skull flat bones, clavicle) directly from mesenchyme, and endochondral (most other bones) from a hyaline cartilage model. Lengthening occurs at the epiphyseal plate until it closes in late adolescence. Throughout life, remodeling continues. PTH raises blood calcium by activating osteoclasts; calcitonin opposes it (modest effect). Estrogen restrains osteoclasts; this is why postmenopausal women lose bone rapidly.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw a long bone in cross-section and label five structures. They have to commit before checking.

Common misconceptions to address

- Bone is thought of as inert. Bone is intensely metabolically active and continuously remodeled.
- Osteoclast and osteoblast names sound similar; students mix them up. Cue: clast like clast-rophobic = breaking down; blast like blast off = building.
- Calcium balance and bone density are sometimes seen as the same thing. They are linked but distinct: short-term calcium balance can be at the expense of bone density.

Order of operations (what to teach first)

Cells, then compact vs spongy structure, then ossification types, then lifelong remodeling and hormonal control.

Self-test prompts (for students before the recall deck)

- Cue yourself: what does each bone cell do?
- How does estrogen affect bone density and why does postmenopausal osteoporosis happen?
- What is happening at the epiphyseal plate and why does it close in adulthood?

Why This Matters to Your Patients

Osteoporosis is bone loss from accelerated osteoclast activity relative to osteoblasts. Postmenopausal women are at high risk because estrogen normally restrains osteoclasts. Bisphosphonates are first-line treatment; they inhibit osteoclasts. Hyperparathyroidism raises blood calcium by mobilizing bone calcium, producing kidney stones, fractures, and 'stones, bones, groans' on the boards. Pediatric fracture management depends on whether the epiphyseal plate is involved (Salter-Harris classification); plate injuries can disrupt growth. Bone metastases are common in advanced cancers and cause pain, fractures, and hypercalcemia. Nurses caring for elderly patients are constantly thinking about fall risk, fracture prevention, and bone density.

TOPIC 2 OF 4 . WEEK 6 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Axial Skeleton

Skull, vertebral column, and thoracic cage: the central axis of the body.

The Science

The axial skeleton is the central line: skull, vertebral column, and thoracic cage. Its job is protection of the most vulnerable organs (brain, spinal cord, heart, lungs) and anchorage for the appendicular skeleton.

Skull: eight cranial bones (frontal, parietal pair, temporal pair, occipital, sphenoid, ethmoid) and fourteen facial bones (maxillae, zygomatics, mandible, and the smaller facial bones). Sutures are immovable joints. The vertebral column has five regions: cervical (7, including the atlas and axis), thoracic (12, articulating with ribs), lumbar (5, the weight-bearers), sacrum (5 fused), coccyx (3-4 fused).

Thoracic cage: sternum (manubrium, body, xiphoid) and 12 pairs of ribs (true 1-7, false 8-10, floating 11-12). Function: protect heart and lungs, anchor breathing muscles. The shape and mobility of the cage are critical for ventilation; chronic obstructive lung disease patients develop a barrel chest because their inflated lungs remodel the cage.

Teaching This Topic

Before the video (drawing prompt)

Ask students to list the five regions of the vertebral column and the count of vertebrae in each, from memory.

Common misconceptions to address

- Students think all vertebrae look the same. Each region has distinctive shape because of different function: cervical for mobility, thoracic for rib articulation, lumbar for weight-bearing.
- Floating ribs are thought of as defects. They are normal anatomy.
- The sacrum is treated as a single bone; it is five fused vertebrae and the fusion is functionally important.

Order of operations (what to teach first)

Skull, then vertebral column with regional contrasts, then thoracic cage and its ventilatory role.

Self-test prompts (for students before the recall deck)

- Without notes, write the number of vertebrae in each region.
- What is unique about C1 (atlas) and C2 (axis) and why?
- How does a compression fracture of a thoracic vertebra change posture and breathing?

Why This Matters to Your Patients

Compression fractures of thoracic and lumbar vertebrae are common in osteoporosis and produce loss of height, kyphosis ('dowager's hump'), and chronic pain. Rib fractures limit ventilation because breathing hurts; nurses watch for atelectasis and pneumonia. Skull fractures depend on where they are: a basilar skull fracture can damage cranial nerves; a temporal bone fracture can rupture the middle meningeal artery and cause an epidural hematoma. Spinal cord injuries are classified by level: a C5 transection preserves diaphragm function but paralyzes the arms and legs; C3 or above paralyzes the diaphragm and the patient needs ventilator support.

TOPIC 3 OF 4 . WEEK 6 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Appendicular Skeleton

Pectoral girdle and upper limb, pelvic girdle and lower limb.

The Science

The appendicular skeleton is everything that hangs off the axial. Pectoral girdle (clavicle and scapula) attaches the upper limb. Pelvic girdle (two coxal bones plus the sacrum) attaches the lower limb. The trade-off across the two girdles is instructive: pectoral is loose (mobility), pelvic is fused (stability).

Upper limb: humerus (arm), radius (lateral, thumb side) and ulna (medial) (forearm), 8 carpals (wrist), 5 metacarpals (palm), 14 phalanges (fingers). Lower limb: femur (thigh, largest bone in the body), tibia (medial weight-bearer) and fibula (lateral) (leg), 7 tarsals (ankle and back of foot), 5 metatarsals (foot), 14 phalanges (toes).

Pelvic girdle has key sex differences: female pelvis is wider, shallower, with a larger pelvic inlet, designed for childbirth. The pubic angle is wider in females (about 100 degrees) versus narrower in males (about 60 degrees). These are clinically relevant in obstetrics, pelvic surgery, and forensic anthropology.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw the bones of one arm and one leg from memory, labeling. They have to commit before correction.

Common misconceptions to address

- Students confuse radius and ulna. Cue: radius is on the same side as the radial artery (thumb side).
- Tibia and fibula get mixed up. Tibia is the weight-bearer (much thicker); fibula is the lateral splint.
- Students assume male and female skeletons are identical. The pelvis differs significantly.

Order of operations (what to teach first)

Pectoral girdle, upper limb proximal to distal, then pelvic girdle, then lower limb. Compare girdle mobility versus stability as a closing concept.

Self-test prompts (for students before the recall deck)

- Which forearm bone is on the thumb side?
- How many bones in the wrist?
- How does the female pelvis differ from the male and why?

Why This Matters to Your Patients

Clavicle fractures are common (the most common fracture in children) and usually heal without surgery, but the shoulder drops because the strut is broken. Hip fractures in the elderly are a major source of morbidity;

mortality at one year is around 20-30%. Carpal tunnel syndrome (median nerve compression at the wrist) is a workplace injury epidemic. Pelvic fractures from high-energy trauma can hide enormous blood loss in the pelvic cavity. Obstetric nursing depends on knowing pelvic anatomy in detail.

TOPIC 4 OF 4 . WEEK 6 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Joints and Body Movements

Classifying joints by structure and function, plus the vocabulary of movement.

The Science

Joints classify two ways. Structurally: fibrous, cartilaginous, synovial, based on what connects the bones. Functionally: synarthrosis (immovable), amphiarthrosis (slightly movable), diarthrosis (freely movable). Most synovial joints are diarthroses; most fibrous and many cartilaginous joints are synarthroses or amphiarthroses.

Synovial joints have the most clinical relevance. They have a joint cavity filled with synovial fluid, articular cartilage on bone ends, a fibrous capsule, and often ligaments, menisci, and bursae. Subtypes: plane (carpals), hinge (elbow, knee, interphalangeal), pivot (atlanto-axial), condyloid (wrist), saddle (thumb), ball-and-socket (shoulder, hip). Each subtype has characteristic movements.

Movements: flexion (decreases joint angle), extension (increases it), abduction (moves away from midline), adduction (moves toward midline), rotation (around long axis), circumduction (cone-shaped sweep), plus the specialty terms (pronation, supination, inversion, eversion, dorsiflexion, plantarflexion). Teach the major terms with both the verbal definition and a body demonstration.

Teaching This Topic

Before the video (drawing prompt)

Have students perform each movement (flexion, extension, abduction, etc.) on themselves and write the joint that allows it. Movement memory is stronger than verbal memory.

Common misconceptions to address

- Students confuse pronation and supination. Cue: supination = palm up, like carrying SOUP.
- Flexion is sometimes thought to mean 'curling.' It means decreasing the joint angle, which is curling for some joints but not for others (the shoulder flexes by raising the arm forward).
- Hinge joints are thought to be limited to one axis. They are, but with some rotation in the knee (which is unusual).

Order of operations (what to teach first)

Classification (structural and functional), then synovial subtypes with examples, then movements with physical demonstration.

Self-test prompts (for students before the recall deck)

- Name the six synovial joint subtypes and give an example of each.
- What is the difference between abduction and adduction?
- Demonstrate flexion and extension of the elbow, hip, and shoulder.

Why This Matters to Your Patients

Rheumatoid arthritis attacks synovial joints; the inflamed synovium forms a pannus that erodes cartilage, then bone, leaving deformed joints. Osteoarthritis is wear and tear: cartilage thins, joint space narrows, osteophytes form. Joint dislocations are more common at the shoulder than the hip because the shoulder trades stability for mobility. Ligament injuries (ACL, MCL of the knee) are common in athletes. Hip replacements and knee replacements are among the most common surgeries in the elderly population. Knowing joint anatomy is part of every assessment of a patient who fell, had a sports injury, or has chronic joint pain.